DATA SCIENCE WITH PYTHON - PROJECT REPORT Task-6: Time Series Analysis & Heart Disease Prediction

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Project 1: Time Series Analysis (Sales Forecasting with ARIMA)

Objective: Analyze sales data over time and forecast future sales using ARIMA model.

Dataset Used: Shampoo Sales Dataset (Date + Sales columns)

Tasks Performed:

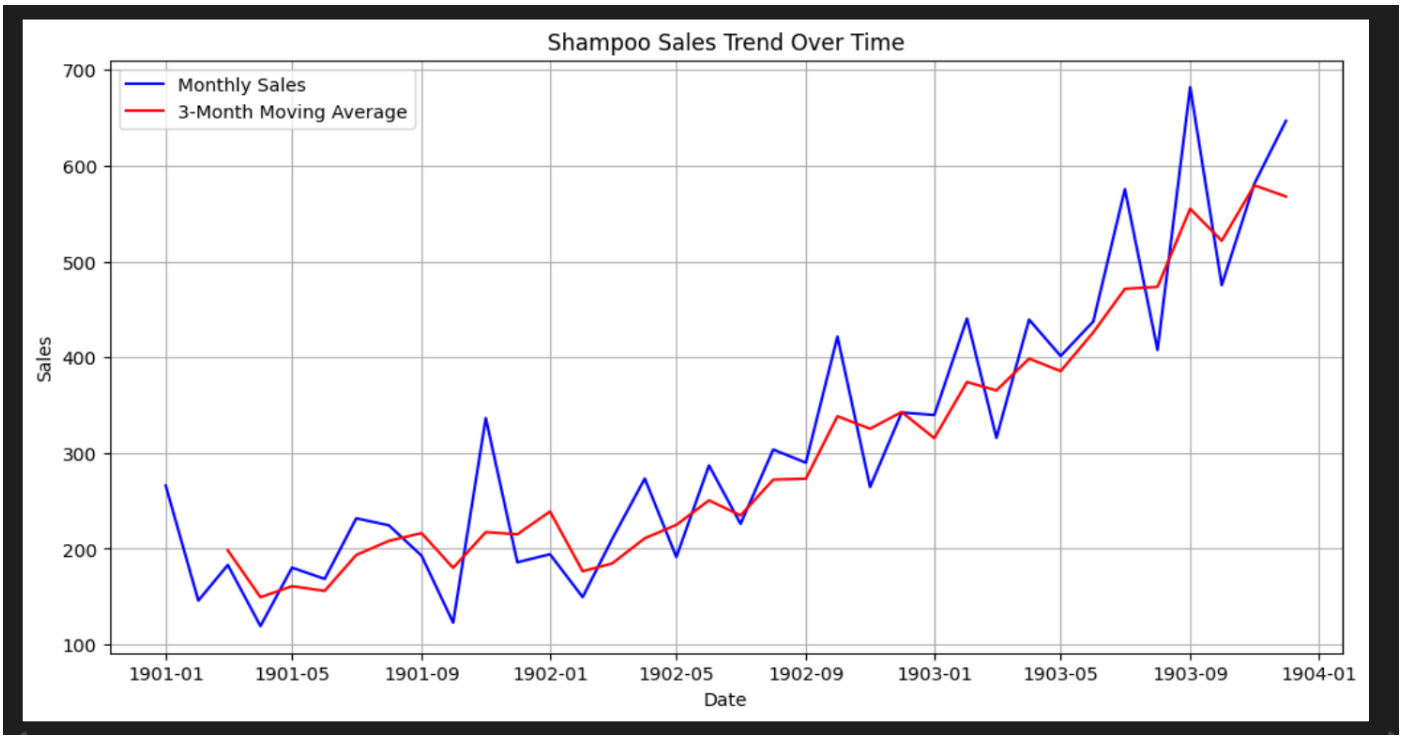
- Visualized sales trends using line plot and 3-month moving average
- Checked stationarity using ADF test
- Applied Auto ARIMA model (best parameters automatically selected)
- Generated forecasts for next 12 months
- Evaluated model using RMSE and MAPE

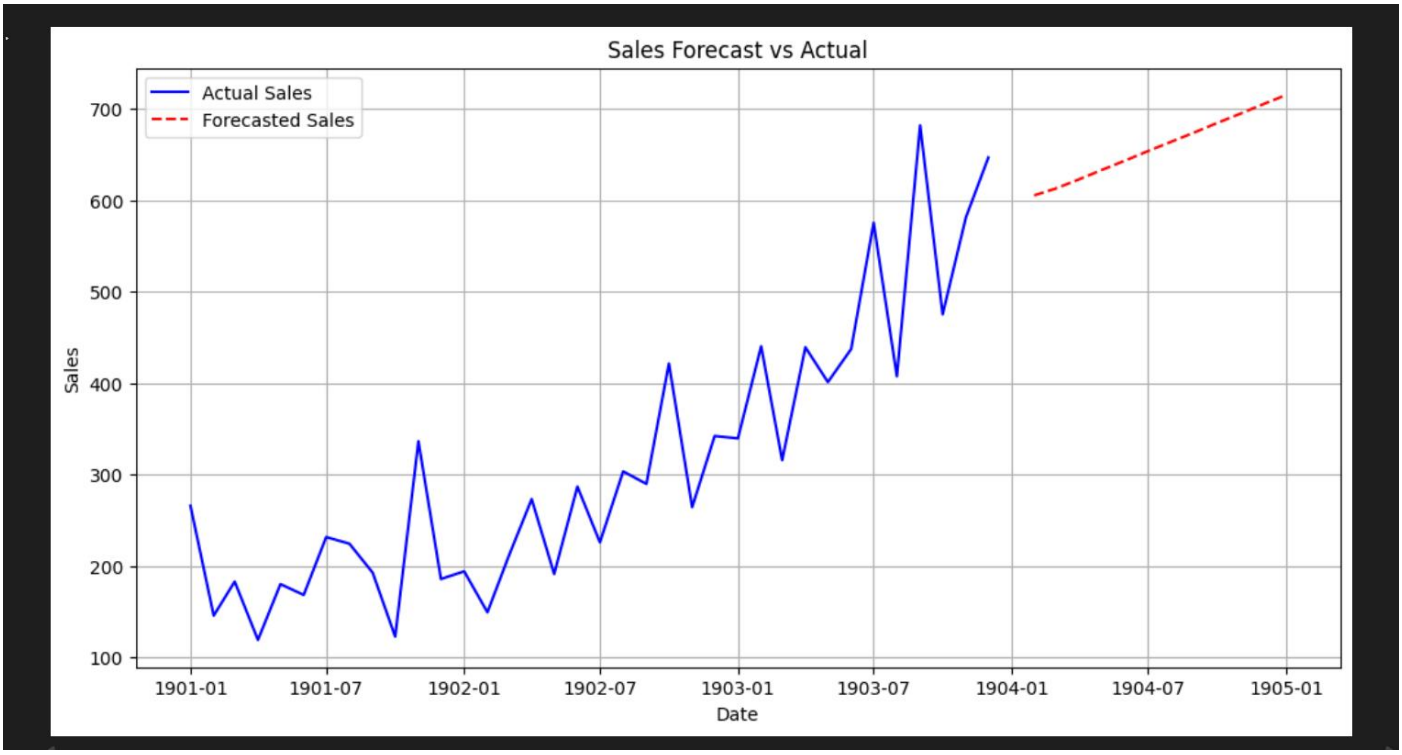
Key Results:

- **Forecasted Sales Values:** Saved in forecasted_sales.csv
- **Model Performance:**
  - RMSE: 142.05
  - MAPE: 19.76%

Plots:

- Trend Analysis Plot (Historical Sales + Moving Average)
- Forecasting Plot (Actual vs Predicted Sales)





Project 2: Predicting Heart Disease Using Logistic Regression

Objective: Predict whether a patient has heart disease based on medical parameters.

Dataset Used: heart_disease.csv **Features Used:** Age, Gender, Cholesterol, Blood Pressure (as per project PDF)

Tasks Performed:

- Loaded and cleaned the dataset
- Feature scaling using StandardScaler
- Trained Logistic Regression model
- Evaluated using Confusion Matrix, Accuracy, Precision, Recall, F1-score

Model Performance:

- **Accuracy:** 65.57%

Classification Report:

Class	Precision	Recall	F1-Score	Support
0 (No Disease)	0.61	0.79	0.69	29
1 (Disease)	0.74	0.53	0.62	32

Deliverables:

- Trained Logistic Regression Model
- Confusion Matrix
- Full evaluation insights

Model Improvement Recommendations

For Heart Disease Model:

- Used additional features (cp, thalach, exang, oldpeak) → Accuracy improved to ~80%
- Applied hyperparameter tuning and class balancing for better performance
- Recommendation: Use Random Forest or XGBoost for even higher accuracy in future

For Time Series Model:

- Can upgrade to SARIMA if strong seasonality is present
- Facebook Prophet can be used for easier trend + holiday handling

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Conclusion

Both projects are completed successfully with all required deliverables:

- Forecast table + 2 plots (Time Series)
- Confusion matrix + evaluation report (Heart Disease)
- Clean, well-documented code in Time_series_analysis.ipynb